

0/1 Knapsack using Recursion & DP (top-down)

$W=3$

profit = [1, 2, 3]

weight = [4, 5, 1]

For DP

$dp[1][3] = 0$

$dp[2][3] = 0$

$dp[2, 4] = 1$

$dp[3][4] = 3$

$KP(3, 4) \max(3, 1) = 3$

include 3rd item

$KP(2, 3) \max(0, 3) = 3$

Can't include because $w > 3$
 $5 > 3$

$0 + 0 = 0$

$KP(1, 3)$

include 1

exclude 1

Can't include item 1 weight $4 > 3$

$KP(0, 3)$

return 0

Profit = 3

Profit = 1

$\max(3, 1) = 3$
so Profit = 3.

exclude 3rd item

$0 + 1 = 1$
 $KP(2, 4)$

include 2

exclude 2

Since weight $5 > 4$

$\max(0, 1) = 1$
 $KP(1, 4)$

include 1

exclude 1

$KP(0, 0)$

$KP(0, 4)$

return '0'

return '0'

$\max(1, 0) = 1$